Quick Work

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M T W T F S S

Tutorial - 4.

Study Hour	Attendence	Assignment	Internal Score	Result
12	85	80	35	1
6	60	55	22	0
15	90	88	40	1
5	58	50	20	0
10	75	70	30	1
8	65	60	25	0
18	95	92	45	1
7	62	58	24	0
14	88	85	38	1

Q find the equation for Logistic Regression.

$$P(Y=1) = \frac{1}{1+e^{-z}} \qquad z = x_1 w_1 + x_2 w_2 + b$$

$w_1 = 0,\ w_2 = 0,\ w_3 = 0,\ w_4 = 0$

$b = 0,\ x_1 = 12,\ x_2 = 85,\ x_3 = 80,\ x_4 = 35.$

$$z_1 = 0. \quad \hat{y}_1 = \sigma(z_1) = \frac{1}{1+e^{0}} = \frac{1}{2} = 0.5.$$

$z_2 = w_1 = 0.1,\ w_2 = 0.1,\ w_3 = 0.1,\ w_4 = 0.1,\ b = 0.$

$z_2 = (6 \times 0.1) + (60 \times 0.1) + (55 \times 0.1) + (22 \times 0.1) + 0 = 35.8.$

$$\hat{y}_2 = \sigma(z_2) = \frac{1}{e^{-35.8}} = 3.52$$

$w_1 = 0.2.$

$z_3 = (15 \times 0.2) + (90 \times 0.2) + (88 \times 0.2) + (40 \times 0.2) + 0 = 44.6$

$$\hat{y}_3 = \sigma(z_3) = \frac{1}{e^{-44.6}} = 2.34$$

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